

Serie 14

INVERSES, DUAL VECTOR SPACE

This exercise sheet is to be handed in the week before the Frühjahr Semester 2026.

We repeat here the first exercise from Sheet 13 as the techniques to solve it were only partially covered in the lecture.

1. Inverses. For each of the following matrices, determine whether or not it is invertible, and if it is compute its inverse.

(a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 2 & -3 & 1 \\ -1 & 3 & -3 & -2 \\ 2 & 0 & 1 & 5 \\ 3 & 1 & -2 & 6 \end{pmatrix}$

2. Dual and bidual. Let $(v_1, \dots, v_n)$ be an ordered basis of a vector space V , let $(v_1^*, \dots, v_n^*)$ the corresponding dual basis of the dual vector space V^* , and let $(v_1^{**}, \dots, v_n^{**})$ the dual basis corresponding to B^* of the bidual vector space $(V^*)^*$. Show, that the natural isomorphism

$$\tau : V \xrightarrow{\sim} (V^*)^*, \quad v \mapsto \tau(v)$$

maps every v_j to the corresponding v_j^{**} .

3. Dual and Hom. Let V and W be finite-dimensional vector spaces over a field K . Show that:

- (a) $\text{Hom}(V, W) \cong \text{Hom}(W^*, V^*)$;
- (b) The following map is an isomorphism

$$\begin{array}{ccc} \text{Hom}(V, V) & \rightarrow & \text{Hom}(V, V)^* \\ T & \mapsto & [S \mapsto \text{tr}(S \circ T)], \end{array}$$

where $\text{tr} \in \text{Hom}(V, V)^*$ is the *trace map* defined by

$$\text{tr}(T) = \text{tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \sum_{i=1}^n ([T]_{\mathcal{B}}^{\mathcal{B}})_{ii},$$

for any basis $\mathcal{B}$ of V .

Remark. You may use this in the above exercise, but as a bonus you could first show that the map $T \in \text{Hom}(V, V) \mapsto \text{tr}(T)$ is independent of the choice of basis of V .

Single Choice. In each exercise, exactly one answer is correct.

1. Let $\dim V = 4$. Then there exists $\varphi \in V^*$ with $\dim \text{Ker } \varphi = 2$.
 - (a) Correct
 - (b) False
2. Every finite dimensional vector space is the dual of another finite dimensional vector space.
 - (a) Correct
 - (b) False
3. The set of all invertible $n \times n$ -matrices is...
 - (a) not a real linear subspace of $M_n(\mathbb{R})$
 - (b) a real linear subspace of $M_n(\mathbb{R})$
4. Let $f : V \rightarrow W$ be an arbitrary homomorphism between two K -vector spaces. Which of the following five assertions is not equivalent to the others?
 - (a) f is injective.
 - (b) The dual map $f^* : W^* \rightarrow V^*$ is surjective.
 - (c) The zero element of V is the only element mapped to the zero element of W .
 - (d) There exists a Homomorphism $g : W \rightarrow V$ with $f \circ g = \text{id}_W$.
 - (e) For every $v \in V \setminus \{0\}$ there exists $\ell \in W^*$ with $\ell(f(v)) \neq 0$.
 - (f) All five assertions are equivalent.

Multiple Choice.

1. For what value of parameter x is matrix $A = \begin{pmatrix} 1 & x & 1 \\ 3 & 3 & x \\ 0 & 3 & 1 \end{pmatrix}$ not invertible?

(a) 0

(b) 1

(c) 2

(d) 3

(e) 4